

THE RESTRICTED HILL FOUR-BODY PROBLEM WITH APPLICATIONS TO THE EARTH–MOON–SUN SYSTEM

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Abstract. Starting from the four-body problem a generalization of both the restricted three-body problem and the Hill three-body problem is derived. The model is time periodic and contains two parameters: the mass ratio ν of the restricted three-body problem and the period parameter m of the Hill Variation orbit. In the proper coordinate frames the restricted three-body problem is recovered as $m \rightarrow 0$ and the classical Hill three-body problem is recovered as $\nu \rightarrow 0$. This model also predicts motions described by earlier researchers using specific models of the Earth–Moon–Sun system. An application of the current model to the motion of a spacecraft in the Sun perturbed Earth–Moon system is made using Hill's Variation orbit for the motion of the Earth–Moon system. The model is general enough to apply to the motion of an infinitesimal mass under the influence of any two primaries which orbit a larger mass.

Using the model, numerical investigations of the structure of motions around the geometric position of the triangular Lagrange points are performed. Values of the parameter ν range in the neighborhood of the Earth–Moon value as the parameter m increases from 0 to 0.195, at which point the Hill Variation orbit becomes unstable. Two families of planar periodic orbits are studied in detail as the parameters m and ν vary. These families contain stable and unstable members in the plane and all have the out-of-plane stability. The stable and unstable manifolds of the unstable periodic orbits are computed and found to be trapped in a geometric area of phase space over long periods of time for ranges of the parameter values including the Earth–Moon–Sun system.

This model is derived from the general four-body problem by rigorous application of the Hill and restricted approximations. The validity of the Hill approximation is discussed in light of the actual geometry of the Earth–Moon–Sun system.

Key words: Hill four-body problem, Earth–Moon–Sun system.

1. Introduction

In this paper a generalization of the restricted three-body problem and the Hill three-body problem is derived and investigated. The equations of motion are self-consistent in that the motion of all the bodies are governed by mutually consistent force laws. The equations contain two parameters and are time periodic. The prime application of this model is to spacecraft motion in the Earth–Moon–Sun system, yet the equations are general enough to apply to a wide variety of problems.

The equations are investigated for a range of parameter values close to the Earth–Moon–Sun values. Periodic orbits and the properties of their stable and unstable manifolds are investigated. Additionally, the dynamics exhibited by the current



system match the dynamics seen in earlier analyses made using more complex models of the specific Earth–Moon–Sun system.

2. The Four-Body Problem

Begin with the general four-body problem, stated in terms of normalized variables. The equations of motion are given in a vector formulation with respect to a rotating coordinate system with constant angular velocity $\mathbf{\Omega}$. These equations govern the motion of the four bodies M_0, M_1, M_2, M_3 with respective mass m_0, m_1, m_2, m_3 :

$$\ddot{\mathbf{R}}_i + \mathbf{\Omega} \times [2\dot{\mathbf{R}}_i + \mathbf{\Omega} \times \mathbf{R}_i] = \frac{1}{m_i} \nabla_{\mathbf{R}_i} U, \quad i = 0, 1, 2, 3 \quad (1)$$

$$U = \sum_{0 \leq i < j \leq 3} \frac{m_i m_j}{|\mathbf{R}_{ij}|}, \quad \mathbf{R}_{ij} = \mathbf{R}_j - \mathbf{R}_i. \quad (2)$$

$\mathbf{R}_i$ is the position vector of the body M_i with respect to the system center of mass, $\dot{\mathbf{R}}_i$ is the time derivative of $\mathbf{R}_i$ with respect to the rotating coordinate frame, the time is scaled so that the gravitational constant is unity, the length is dimensionless with the unit length still to be chosen and the masses are normalized with respect to the mass of M_0 . The following properties hold:

$$m_0 = 1, \quad (3)$$

$$m_j = \mu \alpha_j, \quad j = 1, 2, 3 \quad (4)$$

where the condition and relation exist:

$$\sum_{j=1}^3 \alpha_j = 1, \quad (5)$$

$$\sum_{i=0}^3 m_i = 1 + \mu. \quad (6)$$

These equations possess the classical integrals of angular momentum, linear momentum and energy. As the coordinate origin is co-located with the center of mass of the four bodies there is the additional relationship

$$\sum_{i=0}^3 m_i \mathbf{R}_i = 0. \quad (7)$$

Of interest is the dynamics of the system with respect to the center of mass of the three bodies M_1, M_2 and M_3 , called $\mathbf{R}_c$ and defined by

$$\mathbf{R}_c = \frac{1}{\mu} \sum_{j=1}^3 m_j \mathbf{R}_j. \quad (8)$$

Note immediately that the vectors $\mathbf{R}_c$ and $\mathbf{R}_0$ are directly related as

$$\mathbf{R}_0 = -\mu\mathbf{R}_c. \quad (9)$$

To shift the equations of motion to $\mathbf{R}_c$, define the vector transformation

$$\mathbf{R}_j = \mathbf{R}_c + \mathbf{r}_j, \quad j = 1, 2, 3 \quad (10)$$

where the vector $\mathbf{r}_j$ is the new position vector of M_j with respect to the center of mass of the three bodies. Note the result

$$\sum_{j=1}^3 m_j \mathbf{r}_j = 0. \quad (11)$$

Applying the transformation yields new equations of motion:

$$\dot{\mathbf{R}}_c + \boldsymbol{\Omega} \times [2\dot{\mathbf{R}}_c + \boldsymbol{\Omega} \times \mathbf{R}_c] = - \sum_{k=1}^3 \alpha_k \frac{(1 + \mu)\mathbf{R}_c + \mathbf{r}_k}{|(1 + \mu)\mathbf{R}_c + \mathbf{r}_k|^3}, \quad (12)$$

$$\begin{aligned} \ddot{\mathbf{r}}_i + \boldsymbol{\Omega} \times [2\dot{\mathbf{r}}_i + \boldsymbol{\Omega} \times \mathbf{r}_i] \\ = \sum_{k=1}^3 \alpha_k \frac{(1 + \mu)\mathbf{R}_c + \mathbf{r}_k}{|(1 + \mu)\mathbf{R}_c + \mathbf{r}_k|^3} - \\ - \frac{(1 + \mu)\mathbf{R}_c + \mathbf{r}_i}{|(1 + \mu)\mathbf{R}_c + \mathbf{r}_i|^3} + \mu \sum_{k=1}^3 \alpha_k \frac{\mathbf{r}_{ik}}{|\mathbf{r}_{ik}|^3}, \quad i = 1, 2, 3 \end{aligned} \quad (13)$$

where $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$, the α_i were defined previously and where $\sum_{k=1}^3 = \sum_{k=1, k \neq i}^3$.

3. The Hill Approximation

In its most fundamental form, Hill's approximation entails three assumptions which occur fairly often in physical systems [2]. In terms of the current system these assumptions are

$$\mu \ll 1, \quad (14)$$

$$|\mathbf{r}_{ij}| \ll 1, \quad (15)$$

$$|\mathbf{R}_0 - \mathbf{R}_c| = \mathcal{O}(1). \quad (16)$$

These conditions stipulate that the mass of the body M_0 dominates the system, the bodies M_1 , M_2 and M_3 are close to each other and their center of mass is a finite distance away from the body M_0 . Note that $|\mathbf{R}_0 - \mathbf{R}_c| = (1 + \mu)|\mathbf{R}_c|$ and hence

the restriction is that $|\mathbf{R}_c| = \mathcal{O}(1)$. Condition (15) is made explicit by making the change of coordinates

$$\mathbf{r}_j = \mu^{1/3} \boldsymbol{\rho}_j, \quad j = 1, 2, 3 \quad (17)$$

where $\boldsymbol{\rho}_j$ is a new position vector with respect to $\mathbf{R}_c$. The new coordinates inherit the center of mass relation, Equation (11)

$$\sum_{j=1}^3 \alpha_j \boldsymbol{\rho}_j = 0. \quad (18)$$

Applying Hill's approximation entails introducing these assumptions, expanding the equations of motion in powers of $\mu^{1/3}$ and ignoring orders of $\mu^{2/3}$ and higher. The main expansion used is

$$\begin{aligned} & \frac{(1 + \mu)\mathbf{R}_c + \mu^{1/3}\boldsymbol{\rho}_i}{|(1 + \mu)\mathbf{R}_c + \mu^{1/3}\boldsymbol{\rho}_i|^3} \\ &= \frac{\mathbf{R}_c}{|\mathbf{R}_c|^3} + \frac{\mu^{1/3}}{|\mathbf{R}_c|^3} \boldsymbol{\rho}_i - 3 \frac{\mu^{1/3}\mathbf{R}_c}{|\mathbf{R}_c|^5} (\mathbf{R}_c \cdot \boldsymbol{\rho}_i) + \mathcal{O}(\mu^{2/3}). \end{aligned} \quad (19)$$

To make this expansion valid invoke the assumptions $|\mathbf{R}_c| = \mathcal{O}(1)$ and $\mu \ll 1$.

Introducing this expansion, simplifying using the center of mass relation and dropping orders of $\mu^{2/3}$ and higher, Equations (12) and (13) become

$$\ddot{\mathbf{R}}_c + \boldsymbol{\Omega} \times [2\dot{\mathbf{R}}_c + \boldsymbol{\Omega} \times \mathbf{R}_c] = -\frac{\mathbf{R}_c}{|\mathbf{R}_c|^3} \quad (20)$$

$$\begin{aligned} \ddot{\boldsymbol{\rho}}_i + \boldsymbol{\Omega} \times [2\dot{\boldsymbol{\rho}}_i + \boldsymbol{\Omega} \times \boldsymbol{\rho}_i] &= -\frac{\boldsymbol{\rho}_i}{|\mathbf{R}_c|^3} + 3 \frac{\mathbf{R}_c}{|\mathbf{R}_c|^5} (\mathbf{R}_c \cdot \boldsymbol{\rho}_i) \\ &+ \sum_{k=1}^3 \alpha_k \frac{\boldsymbol{\rho}_{ik}}{|\boldsymbol{\rho}_{ik}|^3}, \quad i = 1, 2, 3. \end{aligned} \quad (21)$$

$\mathbf{R}_c$ can be solved for directly from Equation (20) as it is just the two-body problem cast into a rotating coordinate frame. Once $\mathbf{R}_c$ has been solved for, Equations (21) define a general version of the Hill four-body problem as they involve the mutual attraction of three bodies, all under the additional influence of a fourth body whose motion is known. The three vector Equations (21) may be reduced, via the center of mass relation (18), to two vector equations of motion. Note that the above derivation may be easily extended to the n -body problem.

To solve Equation (20) it is customary to take a solution of the form $\mathbf{R}_c = \text{constant}$ with respect to the rotating coordinate frame, although it is possible to consider an elliptic solution as well. Modify this somewhat and take $\mathbf{R}_c = \mathbf{a} + \mu^{1/3}\mathbf{r}_c$, where $\mathbf{a} = \text{constant}$ with respect to the rotating coordinate frame but where $\mathbf{r}_c$ may be

time varying. Substituting and equating the different orders of $\mu^{1/3}$ in Equation (20) yields

$$\mathbf{\Omega} \times \mathbf{\Omega} \times \mathbf{a} = -\frac{\mathbf{a}}{|\mathbf{a}|^3}, \quad (22)$$

$$\ddot{\mathbf{r}}_c + \mathbf{\Omega} \times [2\dot{\mathbf{r}}_c + \mathbf{\Omega} \times \mathbf{r}_c] = -\frac{\mathbf{r}_c}{|\mathbf{a}|^3} + 3\frac{\mathbf{a}}{|\mathbf{a}|^5} (\mathbf{a} \cdot \mathbf{r}_c). \quad (23)$$

Equation (22) establishes that $\mathbf{a} \cdot \mathbf{\Omega} = 0$. To ensure $|\mathbf{R}_c| = \mathcal{O}(1)$ take $|\mathbf{a}| = 1$, which establishes the unit of length. Then $|\mathbf{\Omega}| = 1$.

Equation (23) then becomes

$$\ddot{\mathbf{r}}_c + \mathbf{\Omega} \times [2\dot{\mathbf{r}}_c + \mathbf{\Omega} \times \mathbf{r}_c] = -\mathbf{r}_c + 3\mathbf{a}(\mathbf{a} \cdot \mathbf{r}_c). \quad (24)$$

This is a linear, time invariant system of differential equations and may be easily integrated once a given realization is introduced. Thus the three-body center of mass is in a ‘near circular’ orbit. The eccentricity of this orbit is of the order of $\mu^{1/3}$.

Returning to the relative equations of motion for the three bodies, Equation (21), $\mathbf{R}_c$ may be replaced with $\mathbf{a}$ to the same order of accuracy as is kept in the equations. Thus, at the lowest order, when the three-body center of mass moves on a near circular orbit, the motion of the three-body center of mass is decoupled from the relative motion of the three bodies. Making this substitution the equations of motion (21) become

$$\begin{aligned} \ddot{\mathbf{p}}_i + \mathbf{\Omega} \times [2\dot{\mathbf{p}}_i + \mathbf{\Omega} \times \mathbf{p}_i] = & -\mathbf{p}_i + 3\mathbf{a}(\mathbf{a} \cdot \mathbf{p}_i) \\ & + \sum_{k=1}^3 \alpha_k \frac{\mathbf{p}_{ik}}{|\mathbf{p}_{ik}|^3}, \quad i = 1, 2, 3 \end{aligned} \quad (25)$$

where $\mathbf{p}_{ik} = \mathbf{p}_k - \mathbf{p}_i$ and with the center of mass relationship (18) remaining.

For current purposes Equations (25) are called the Hill four-body problem. Note that there is a linear momentum and energy integral present for this system. The angular momentum integral does not exist in this set of equations.

4. The Restricted Approximation

Starting from Equations (25), which essentially assume that the center of mass of these bodies is in a circular orbit about the sun, introduce the restricted approximation by taking $\alpha_3 \rightarrow 0$. Then from Equations (5) and (18),

$$\alpha_1 \mathbf{p}_1 + \alpha_2 \mathbf{p}_2 = 0, \quad (26)$$

$$\alpha_1 + \alpha_2 = 1. \quad (27)$$

Introduce a new parameter $\nu \leq 1/2$ and a new vector $\mathbf{p}$ as

$$\alpha_1 = 1 - \nu, \quad (28)$$

$$\alpha_2 = \nu, \quad (29)$$

$$\mathbf{p} = \mathbf{p}_2 - \mathbf{p}_1. \quad (30)$$

Then the equations of motion for M_1 and M_2 , Equation (25) with $i = 1, 2$, do not contain the vector $\mathbf{p}_3$ and thus are independent of M_3 . Eliminate one of the vector equations of motion of M_1 and M_2 using Equation (27) and (30) to arrive at the equation

$$\ddot{\mathbf{p}} + \boldsymbol{\Omega} \times [2\dot{\mathbf{p}} + \boldsymbol{\Omega} \times \mathbf{p}] = -\mathbf{p} + 3\mathbf{a}(\mathbf{a} \cdot \mathbf{p}) - \frac{\mathbf{p}}{|\mathbf{p}|^3}. \quad (31)$$

Once the vector $\mathbf{p}$ is solved for use the following relations to recover the motion of M_1 and M_2 :

$$\mathbf{p}_1 = -\nu\mathbf{p}, \quad (32)$$

$$\mathbf{p}_2 = (1 - \nu)\mathbf{p}. \quad (33)$$

In the following sections we will refer to the bodies M_1 and M_2 as the primaries.

The equation of motion for the body M_3 , Equation (25) with $i = 3$, then becomes

$$\begin{aligned} \ddot{\mathbf{p}}_3 + \boldsymbol{\Omega} \times [2\dot{\mathbf{p}}_3 + \boldsymbol{\Omega} \times \mathbf{p}_3] = & -\mathbf{p}_3 + 3\mathbf{a}(\mathbf{a} \cdot \mathbf{p}_3) - (1 - \nu) \frac{\mathbf{p}_3 + \nu\mathbf{p}}{|\mathbf{p}_3 + \nu\mathbf{p}|^3} \\ & - \nu \frac{\mathbf{p}_3 - (1 - \nu)\mathbf{p}}{|\mathbf{p}_3 - (1 - \nu)\mathbf{p}|^3}. \end{aligned} \quad (34)$$

This equation of motion depends on the solution $\mathbf{p}(t)$ and is, in general, a time varying differential equation. This equation may be defined to be the restricted Hill four-body problem although we will reserve the name for a more specific system. Equation (34) will have the energy integral of motion only in the special case when $\mathbf{p}$ is a stationary solution, which is possible, yet which will not occur in situations of interest.

5. Relative Motion of the Primaries

Now introduce a coordinate realization to express the equations of relative motion of M_1 and M_2 , Equation (31), and of their center of mass motion, Equation (24), in scalar form. Introduce a rotating Cartesian system ($\mathbf{i}, \mathbf{j}, \mathbf{k}$) with the following realizations:

$$\mathbf{a} = \mathbf{i}, \quad (35)$$

$$\mathbf{\Omega} = \mathbf{k}, \quad (36)$$

$$\mathbf{r}_c = x_c \mathbf{i} + y_c \mathbf{j} + z_c \mathbf{k}, \quad (37)$$

$$\mathbf{\rho} = \xi \mathbf{i} + \eta \mathbf{j} + \zeta \mathbf{k}. \quad (38)$$

where $|\mathbf{a}| = a = 1$ and $|\mathbf{\Omega}| = 1$.

Then Equation (24) becomes

$$\begin{aligned} \ddot{x}_c - 2\dot{y}_c &= 3x_c, \\ \ddot{y}_c + 2\dot{x}_c &= 0, \\ \ddot{z}_c &= -z_c. \end{aligned} \quad (39)$$

These are easily integrated [2] to find

$$\begin{aligned} x_c(t) &= C_1 \cos t + C_2 \sin t + C_3, \\ y_c(t) &= 2C_2 \cos t - 2C_1 \sin t - \frac{3}{2}C_3 t + C_4, \\ z_c(t) &= C_5 \cos t + C_6 \sin t, \end{aligned} \quad (40)$$

where the C_i are constants of integration.

The equations of motion for the relative motion of M_1 and M_2 , Equation (31), become

$$\begin{aligned} \ddot{\xi} - 2\dot{\eta} &= 3\xi - \frac{\xi}{(\xi^2 + \eta^2 + \zeta^2)^{3/2}}, \\ \ddot{\eta} + 2\dot{\xi} &= -\frac{\eta}{(\xi^2 + \eta^2 + \zeta^2)^{3/2}}, \\ \ddot{\zeta} &= -\zeta - \frac{\zeta}{(\xi^2 + \eta^2 + \zeta^2)^{3/2}}. \end{aligned} \quad (41)$$

These are known as the Hill equations for the motion of the moon. They are non-integrable and only a few particular solutions are known. Among these known solutions the most important is the ‘Variation’ orbit family which serves as the first approximation in the modern theory of lunar motion. The solution is expressed as a Fourier series and may be found to any desired accuracy. The Variation orbits are periodic orbits with period $2\pi m$ where m is the parameter of the family of periodic orbits. The solution is expressed [7] as

$$\begin{aligned} \xi(t; m) &= \sum_{n=-\infty}^{\infty} a_n(m) \cos \left[\frac{(2n+1)t}{m} \right], \\ \eta(t; m) &= \sum_{n=-\infty}^{\infty} a_n(m) \sin \left[\frac{(2n+1)t}{m} \right], \\ \zeta(t; m) &= 0, \end{aligned} \quad (42)$$

where the values of the coefficients $a_n(m)$ are given to order m^6 in the appendix.

Members of this family are stable if they are retrograde orbits ($m < 0$) and are stable for direct orbits in the interval $0 < m < 0.19510486 \dots$ [1].

6. The Restricted Hill Four-Body Problem

Substituting solution (42) into Equations (34) for M_3 , they become time periodic differential equations with period $2\pi m$ and contain two free parameters, ν and m . If $\nu \rightarrow 0$ in Equations (34), note that the classical Hill equations of motion are recovered.

Now introduce some coordinate changes to put the equations of motion (34) in a more convenient form for the current investigations. First, transform to a coordinate system rotating at an absolute rate $(1 + 1/m)$ in the $\mathbf{\Omega}$ direction. Then define a new position vector $\mathbf{r}$ and a new time parameter τ via

$$\mathbf{\rho}_3(t) = a_0(m)\mathbf{r}\left(\frac{t}{m}\right), \quad (43)$$

$$t = m\tau, \quad (44)$$

where

$$a_0(m) = m^{2/3} \left(1 - \frac{2}{3}m + \frac{7}{18}m^2 - \frac{4}{81}m^3 + \dots\right). \quad (45)$$

The equations of motion (34) can then be reduced to

$$\begin{aligned} \mathbf{r}'' + (1 + m)\mathbf{\Omega} \times [2\mathbf{r}' + (1 + m)\mathbf{\Omega} \times \mathbf{r}] \\ = -m^2\mathbf{r} + 3m^2\mathbf{a}(\mathbf{a} \cdot \mathbf{r}) - \frac{m^2}{a_0^3}(1 - \nu) \frac{\mathbf{r} + \nu(\mathbf{\rho}/a_0)}{|\mathbf{r} + \nu(\mathbf{\rho}/a_0)|^3} \\ - \frac{m^2}{a_0^3}\nu \frac{\mathbf{r} - (1 - \nu)(\mathbf{\rho}/a_0)}{|\mathbf{r} - (1 - \nu)(\mathbf{\rho}/a_0)|^3}, \end{aligned} \quad (46)$$

where $\mathbf{r}'$ denotes the derivative with respect to τ in the rotating coordinate frame.

Introduce the Cartesian system $(\mathbf{i}_m, \mathbf{j}_m, \mathbf{k})$ with the following realizations and relations to the $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ system:

$$\mathbf{i} = \cos \tau \mathbf{i}_m - \sin \tau \mathbf{j}_m, \quad (47)$$

$$\mathbf{j} = \sin \tau \mathbf{i}_m + \cos \tau \mathbf{j}_m, \quad (48)$$

$$\mathbf{r} = x\mathbf{i}_m + y\mathbf{j}_m + z\mathbf{k}, \quad (49)$$

$$\mathbf{\Omega} = \mathbf{k}, \quad (50)$$

$$\mathbf{a} = \cos \tau \mathbf{i}_m - \sin \tau \mathbf{j}_m, \quad (51)$$

$$\frac{\mathbf{p}}{a_0} = (1 + \bar{\xi})\mathbf{i}_m + \bar{\eta}\mathbf{j}_m, \quad (52)$$

$$\bar{\xi}(\tau; m) = \sum_{n=1}^{\infty} \left(\frac{a_n(m)}{a_0(m)} + \frac{a_{-n}(m)}{a_0(m)} \right) \cos 2n\tau, \quad (53)$$

$$\bar{\eta}(\tau; m) = \sum_{n=1}^{\infty} \left(\frac{a_n(m)}{a_0(m)} - \frac{a_{-n}(m)}{a_0(m)} \right) \sin 2n\tau. \quad (54)$$

Then, in scalar variables, equations of motion (46) become

$$\begin{aligned} x'' - 2(1+m)y' &= V_x, \\ y'' + 2(1+m)x' &= V_y, \\ z'' &= V_z, \end{aligned} \quad (55)$$

where

$$\begin{aligned} V(x, y, z, \tau; \nu, m) &= \frac{1}{2} (1 + 2m + \frac{3}{2}m^2) (x^2 + y^2) - \frac{1}{2}m^2 z^2 \\ &\quad + \frac{3}{4}m^2 ((x^2 - y^2) \cos 2\tau - 2xy \sin 2\tau) \\ &\quad + \frac{m^2}{a_0^3} \left[\frac{1-\nu}{R_{1-\nu}} + \frac{\nu}{R_\nu} \right] \end{aligned} \quad (56)$$

and where

$$R_{1-\nu} = \sqrt{[x + \nu(1 + \bar{\xi})]^2 + [y + \nu\bar{\eta}]^2 + z^2}, \quad (57)$$

$$R_\nu = \sqrt{[x - (1 - \nu)(1 + \bar{\xi})]^2 + [y - (1 - \nu)\bar{\eta}]^2 + z^2}. \quad (58)$$

These equations are periodic in τ with period π . They contain the two parameters ν and m . If $m \rightarrow 0$ then the equations of motion for the restricted three-body problem are recovered. Thus Equations (55) are a generalization of both the restricted three-body problem and the Hill equations of motion.

The parameter m in these equations has its origin as the period parameter in the Variation orbit, but it has an additional significance here. As noted by Wintner [7], the distance of the primaries in the Variation orbit is on the order of m^2 . Thus by scaling our distance by $a_0^3(m) \approx m^2$ the length in this system is scaled by the distance between the primaries. In the new units the distance of the Sun from the system of the primaries is on the order of $1/m^2$. Thus as $m \rightarrow 0$ the assumption is that the distance of the Sun from the primaries becomes infinite, while the gradient of the Sun's force on the system, which scales as m^2 , goes to zero. Thus the parameter m is related to the relative distance of the Sun from the primaries in this model.

There is an important symmetry in these equations of motion. Consider the transformation

$$(x, y, \tau) \rightarrow (x, -y, -\tau), \quad (59)$$

which implies the additional transformation

$$(x', y', x'', y'') \rightarrow (-x', y', x'', -y'') \quad (60)$$

and note that the equations of motion remain invariant. This symmetry has two important implications which are used later in the paper. First, given any motion in the system starting at time $\tau = \tau_0$ with initial conditions (x_0, y_0, x'_0, y'_0) and ending at $\tau = \tau_f$, using the symmetry property we immediately know the history of the state $(x_0, -y_0, -x'_0, y'_0)$ at $\tau = -\tau_0$ back to the time $\tau = -\tau_f$. This property will prove useful in the discussion of stable and unstable manifolds of unstable periodic orbits.

The other important implication relates to the existence of periodic orbits. Using the above symmetry property, it is easy to show that if at time $\tau = \tau_0$ the initial conditions (x_0, y_0, x'_0, y'_0) lead to a periodic orbit of period $n\pi$ then at $\tau = -\tau_0$ the initial conditions $(x_0, -y_0, -x'_0, y'_0)$ will also lead to a periodic orbit of the same period. If one of the periodic orbits is unstable, then the other is also. Furthermore, we see that the stable and unstable manifolds of one of the orbits becomes the unstable and stable manifolds, respectively, of the other orbit under the transformation.

Equations (55) are convenient for numerical integration, while for analytical work it is preferable to expand the potential (56) in powers of m :

$$\begin{aligned} V(x, y, z, \tau; v, m) &= \left(1 + 2m + \frac{3}{2}m^2\right) V_0(x, y; v) - \frac{1}{2}m^2 z^2 + m^2 \cos(2\tau) \frac{3}{4}(x^2 - y^2) + \\ &\quad + m^2 \cos(2\tau) v(1-v) \left(\frac{x+v}{r_{1-v}^3} - \frac{x-(1-v)}{r_v^3} \right) - \\ &\quad - m^2 \sin(2\tau) \left[\frac{3}{2}xy + \frac{11}{8}v(1-v) \left(\frac{1}{r_{1-v}^3} - \frac{1}{r_v^3} \right) y \right] + \mathcal{O}(m^3), \end{aligned} \quad (61)$$

where

$$r_{1-v} = \sqrt{(x+v)^2 + y^2 + z^2}, \quad (62)$$

$$r_v = \sqrt{(x-(1-v))^2 + y^2 + z^2} \quad (63)$$

and where $V_0(x, y; v)$ is the potential from the restricted three-body problem:

$$V_0(x, y; v) = \frac{1}{2}(x^2 + y^2) + \frac{1-v}{r_{1-v}} + \frac{v}{r_v}. \quad (64)$$

This form of the potential, Equation (61), is more suitable for analytical work, the only restriction being that the body M_3 not come too close to either body M_1 or M_2 , should this occur a different expansion would have to be used.

The most obvious application of these equations of motion is to spacecraft motion in the Earth–Moon–Sun system. Thus it is of interest to point out what has been neglected and what has been accounted for in comparison to this specific system.

The general Hill approximation accounts for the major influence of the sun on the three bodies and allows one to solve for the approximate motion of the center of mass of these bodies as an ellipse. This center of mass solution is then used in the remaining equations of motion to determine the modified equations of motion accounting for the solar influence. We have seen that, to the order of accuracy of the approximation, we may assume that the nominal center of mass motion may be considered circular in these equations of motion if it has an actual eccentricity on the order of $\mu^{1/3}$. For the Earth–Moon–Sun system the actual eccentricity is 0.0167 . . . while $\mu^{1/3} = 0.0145$. . . , so this is a valid assumption for this system.

What has not been properly accounted for is the fact that the Earth–Moon ecliptic and the Earth–Moon–Sun ecliptic are not coplanar, but are inclined $\approx 5^\circ$ with respect to each other. There are a variety of ways one could introduce this inclination into the problem, the most direct way being to solve for the ζ component in the relative equations of motion for M_1 and M_2 . Another approach would be to assume that the coordinate system angular velocity is of the form $\boldsymbol{\Omega} = \sigma \mathbf{j} + \sqrt{1 - \sigma^2} \mathbf{k}$ where σ represents the angle between the two planes. If we use the latter approach we see that, to the order of the approximation made, we may ignore a difference in angle on the order of $\mu^{1/3}$ radians, or approximately 1° in the Earth–Moon–Sun case. Thus, it is not strictly proper to neglect the inclination between the ecliptic planes. In future investigations the inclusion of out-of-plane effects would be important.

Finally, with respect to the Earth–Moon–Sun system, the Variation orbit only describes the qualitative Earth–Moon motion and the true solution will not match with the nice periodicity exhibited by the Variation orbit. However, the differences are, in general, of magnitude smaller than $\mu^{1/3}$ and thus can be viewed as perturbations to the motion of the primaries and hence perturbations to the dynamics of the spacecraft.

Leaving these valid concerns behind, we now approach the equations of motion as a mathematical problem and attempt to determine solutions of interest and possible regions of phase space where regular motion is possible. We will consider motion in the plane only ($z \equiv 0$). If regular motions are found, we will determine the out-of-plane stability as a matter of course.

7. Application to Motion Near L_4 and L_5

Using the above model, equations of motion (55) with potential (56), we make some elementary observations using numerical analysis to compute families of periodic orbits and their stability properties in a region of parameter space in the vicinity of the L_4 and L_5 equilibrium points of the restricted three body problem. These equilibrium points are often known as the equilateral triangle points as they are located, geometrically, at the vertex of an equilateral triangle with the two primaries

at the base. In our system these points are equilibrium points only for the case $m = 0$, and are not well defined for values of $m > 0$. In the following, however, we will define the natural continuation of these equilibrium points when m increases from zero.

We investigate this area of phase space over a restricted region of the parameter space:

$$0.0115 \leq \nu \leq 0.0130 \quad (65)$$

$$0.0 \leq m \leq 0.195. \quad (66)$$

The ν value corresponding to the Earth–Moon system is ≈ 0.0122 , thus the interval varies around this value by approximately $\pm 6\%$. The range of the m parameter corresponds to no solar influence at $m = 0$ to a maximum influence at $m = 0.195$ when the Variation orbit of the two primaries becomes unstable. The m value of the Earth–Moon–Sun system is ≈ 0.0808 . The parameter region is chosen to concentrate on the Earth–Moon–Sun system.

7.1. PRELIMINARIES

We use the Poincaré map reduction and its attendant simplifications to present and compute our results. A Poincaré map is, in general, defined by the reduction of the dynamical system by one degree of freedom, realized by the restriction of the dynamics to a hyperplane of co-dimension 1 to the phase space. Certain transversality conditions on the trajectory with respect to this hyperplane must be enforced, in general. In our case, we have a time-periodic system meaning that the natural coordinate for the Poincaré map is the time coordinate. We construct the map by choosing an initial time τ_0 and consider the dynamics of the system at intervals $\tau_0 + n\pi$ where $n = 0, \pm 1, \pm 2, \dots$, where we arbitrarily set $\tau_0 = 0$. This is also known as the stroboscopic map since it is applied to the time coordinate. Using this representation the dynamical system is described by a four-dimensional, non-linear, area-preserving map. There are no simple invariant surfaces contained in this map as the equations of motion do not have an energy integral except at $m = 0$ where the four-dimensional map is degenerate. Under this map a periodic family of period π is represented as a family of fixed points in the four-dimensional space. Similarly, a periodic family of period 2π is represented as a family of two points in the four-dimensional space, alternating under single iterations of the map.

The current analysis concentrates on orbits associated with the L_4 triangular point (in the positive y region of the coordinate space). Corresponding to every orbit presented there is another orbit found by reversing time and applying the transformation to the initial conditions

$$(x_0, y_0, x'_0, y'_0) \rightarrow (x_0, -y_0, -x'_0, y'_0). \quad (67)$$

Which transforms all orbits in the L_4 region into orbits in the L_5 region.

To evaluate the stability of the periodic orbits we compute the 4×4 monodromy matrix and evaluate its eigenvalues. As the characteristic polynomial of our matrix is reciprocal we used standard techniques to determine the stability of the fixed points [5].

7.2. PERIOD π FAMILY

If we consider the model for $m = 0$ we recall from the restricted three-body problem that there are stable equilibrium points located at the triangular Lagrange points, if $\nu < 0.0385 \dots$ and not equal to $0.0243 \dots$ and $0.0135 \dots$ [5]. It is easy to establish that these equilibrium points will continue into periodic orbits of period π as m increases from 0. To do so note that scaling the time in Equations (55) by $(1 + m)$ and dividing the potential, Equation (61), by the resulting factor $(1 + m)^2$ results in the standard restricted three-body potential plus additional terms of order m^2 and higher. Thus, to first order in m the natural frequencies of the equations are equal to the natural frequencies of the triangular Lagrange points scaled by a factor $(1 + m)$. To ensure that the equilibrium solutions can be extended to the time-periodic case we only need to verify that the natural frequencies will not be in a $1:M$ resonance with the forcing potential, i.e. do not have a period $M\pi$. From Reference [5], page 50, we have the natural frequencies of the equilateral points

$$\omega_1 = \sqrt{\frac{1}{2}[1 - \sqrt{1 - 27\nu + 27\nu^2}]} \quad (68)$$

$$\sim \frac{3}{2}\sqrt{3\nu(1 - \nu)}, \quad (69)$$

$$\omega_2 = \sqrt{\frac{1}{2}[1 + \sqrt{1 - 27\nu + 27\nu^2}]} \quad (70)$$

$$\sim 1 - \frac{27}{8}\nu(1 - \nu), \quad (71)$$

the expansions being performed assuming small values of $\nu(1 - \nu)$. Thus in our system, Equations (55), the natural frequencies become $\omega_1(1 + m)$ and $\omega_2(1 + m)$ plus higher order terms in m , and the non-resonance condition is that neither of them be equal to 2 (as then the time-periodic potential would force the system at its natural frequency, and a more sophisticated analysis would be needed). This condition is easily established for both of these frequencies: $\omega_1(1 + m) \ll 1$ as $\nu(1 - \nu) \ll 1$, and $\omega_2(1 + m) < 2$ as $m \ll 1$.

From this simple analysis we also can develop a limit for when these continued periodic orbits may lose their stability by passage through a 2:1 resonance condition, which will occur when the frequencies pass through the value 1 (which corresponds to the orbit multipliers passing through a value of -1 on the unit circle). Again we note that $\omega_1(1 + m) < 1$ for values of interest, but that $\omega_2(1 + m)$ may equal 1 for small values of m and $\nu(1 - \nu)$. Denoting m_{cr} as the value of m at which this 2:1 resonance is satisfied, we see

$$m_{\text{cr}} \sim \frac{27}{8}\nu(1 - \nu). \quad (72)$$

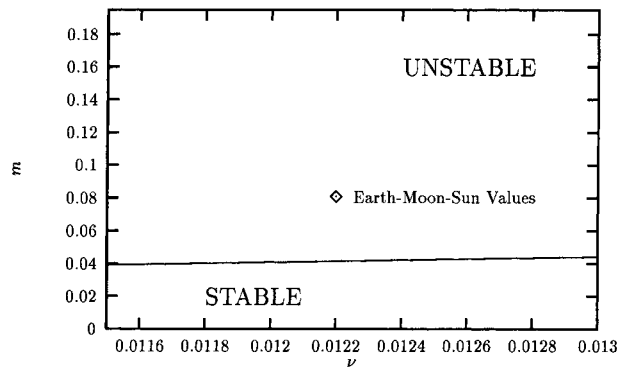


Figure 1. Stable and unstable regions of the ν and m parameter space for period π orbits.

To extend the analysis beyond small values of m we numerically computed the fixed points of the map as m increased from zero for a range of ν values in the parameter space. Figure 1 shows the stability and instability regions of these orbits in the parameter space. The transition between stable and unstable orbits corresponds to passage of the periodic orbit multipliers through the value -1 on the unit circle, as expected from the above analysis. Also, plotting the simple relation for the critical value of m , Equation (72), against the numerically determined values given in Figure 1 yields indistinguishable differences for the resolution of that figure. Note that the instability regions are only singly unstable as there still exist a pair of multipliers on the unit circle. For the Earth–Moon–Sun parameter values the orbit has a period of a half-month and is unstable. All members of this family in the parameter region have the out-of-plane stability.

See Figures 2–4 for a plot of the initial conditions at $\tau = 0$ for a fixed ν as m increases. Figure 3 is a detail of the coordinate space. A few orbits are drawn to show

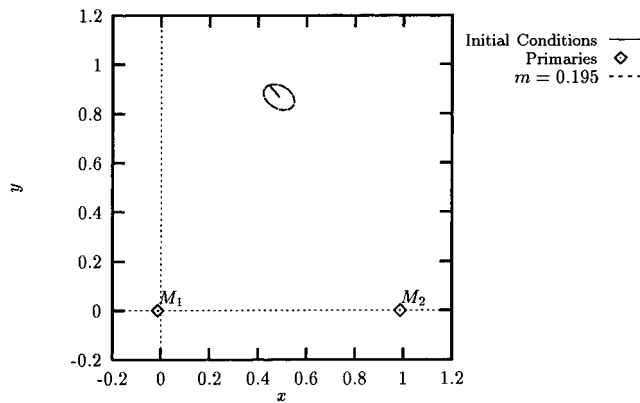


Figure 2. (x, y) initial condition coordinates for period π orbits at $\tau = 0$, $\nu = 0.0122$ as m varies. The periodic orbit trajectory for $m = 0.195$ is also drawn.

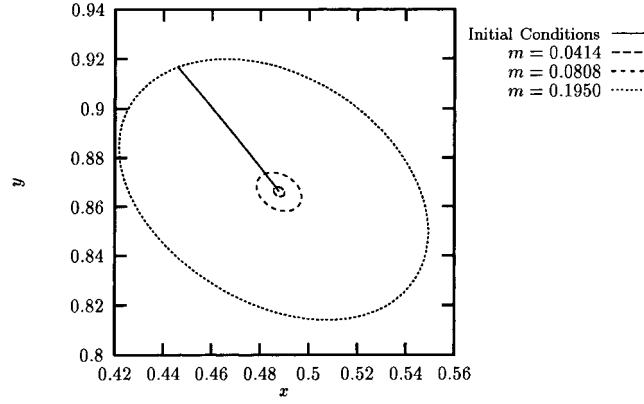


Figure 3. Detail of (x, y) initial condition coordinates for period π orbits at $\tau = 0$, $\nu = 0.0122$ as m varies. The periodic orbit trajectories for $m = 0.0414$, 0.0808 and 0.195 are also drawn.

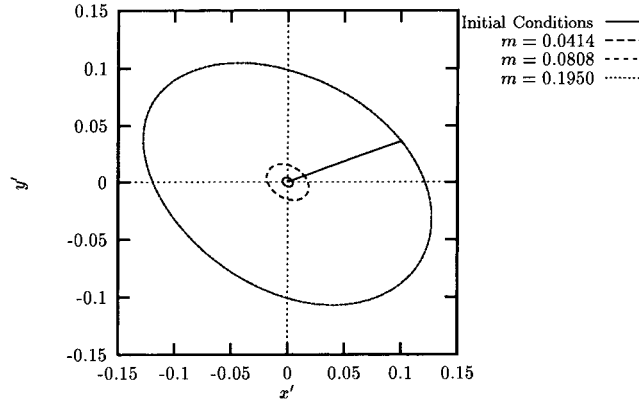


Figure 4. (x', y') initial condition coordinates for period π orbits at $\tau = 0$, $\nu = 0.0122$ as m varies. The periodic orbit velocity hodograph for $m = 0.0414$, 0.0808 and 0.195 are also drawn.

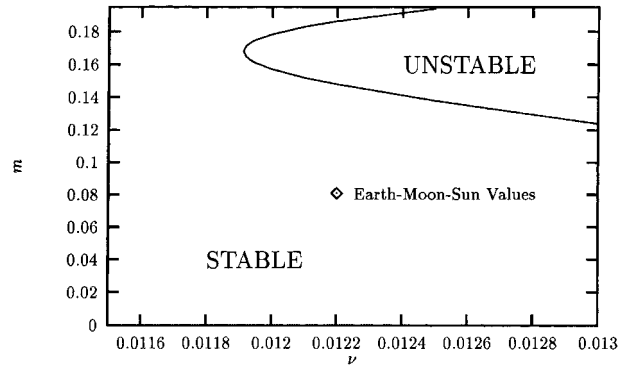


Figure 5. Stable and unstable regions of parameter space ν and m for period 2π orbits.

their shapes. We will refer to these fixed points of the map as the L_4 and L_5 fixed points as they are the natural generalization of the restricted three-body problem triangular fixed points as m is increased from 0.

7.3. PERIOD 2π FAMILY

A family of periodic orbits with period 2π also exist in the vicinity of the L_4 and L_5 points. For most values of m and ν in our parameter space this is a stable family (see Figure 6). See Figures 6 and 7 for the initial conditions at $\tau = 0$ for a fixed ν as m varies. A few orbits have been drawn to show their shapes for different values of m . There are two possible initial conditions for each periodic orbit.

As m becomes small the current family traces out a path that resembles a horseshoe orbit in the restricted three-body problem. Note that there exists a stable family of horseshoe orbits of period 2π for ν values in our range in the restricted three-body problem. Thus, we conjecture that these orbits are related to each other as $m \rightarrow 0$. However, this bifurcation cannot be investigated numerically with our current scheme as the four-dimensional map is degenerate at $m = 0$ due to the existence of the Jacobi integral.

As m increases the current 2π period family retains its stability and the individual orbits have a decreasing amplitude and seem to contract around their respective triangular points. As m grows the trajectories reach a minimal size and afterward grow in size again. At their minimal size the trajectory of the orbit describes, approximately, a 2:1 ellipse with length on the order of the primary distance in the rotating reference frame (see Figures 6 and 7). As the trajectories grow after their minimal size is reached, family members with $\nu > 0.01191$ lose stability by passing through the -1 multiplier. Some of these members (for $\nu < 0.01250$) regain stability again before the maximum value $m = 0.195$ is reached. All members of this family have the out-of-plane stability. For parameter values near the Earth–Moon–Sun values this

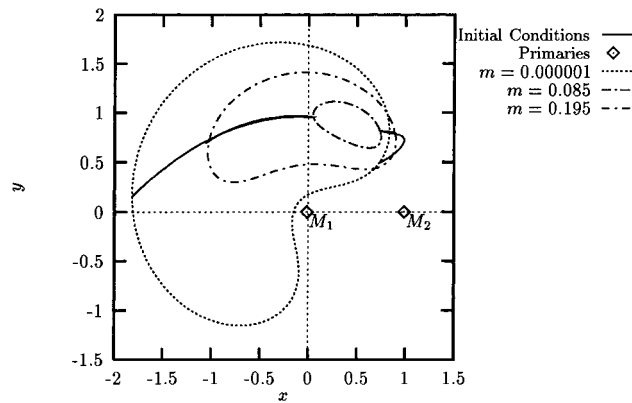


Figure 6. (x, y) initial condition coordinates for period 2π orbits at $\tau = 0, \pi$, $\nu = 0.0122$ as m varies. The periodic orbit trajectories for $m = 1 \times 10^{-6}$, 0.085 and 0.195 are also drawn.

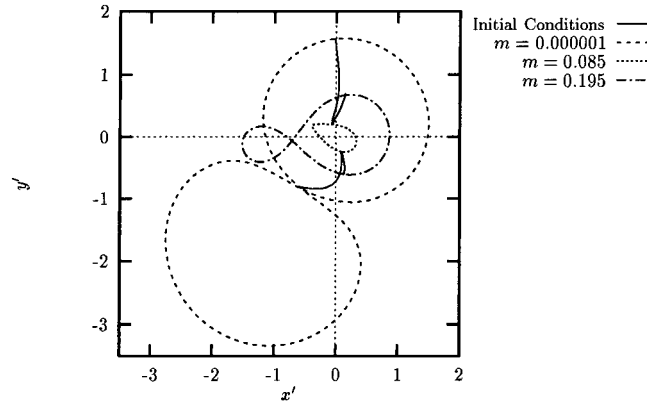


Figure 7. (x', y') initial condition coordinates for period 2π orbits at $\tau = 0, \pi$, $\nu = 0.0122$ as m varies. The periodic orbit velocity hodographs for $m = 1 \times 10^{-6}$, 0.085 and 0.195 are also drawn.

periodic orbit is stable and has a period of one month. See Figure 5 for a plot of the stability and instability regions in the parameter space. Again, all unstable orbits are singly unstable.

At the stability bifurcation in the parameter space a stable family of periodic orbits of period 4π born. If one fixes ν and varies m the 4π family is born when the 2π family loses stability and terminates when the 2π family regains stability.

For values of ν outside the parameter region additional bifurcations of increased complexity occur. More precisely, for larger ν values the instability region grows, taking over a large portion of the parameter space. For smaller ν values the period 2π family may terminate on the period π family, causing the period π family to become stable.

7.4. STABLE AND UNSTABLE MANIFOLDS

An interesting result has been found concerning the stable and unstable manifolds of the unstable period π orbits discussed here. For values of m small enough (< 0.11) the stable and unstable manifolds of the unstable members of this family are trapped in a finite region of phase space for, seemingly, arbitrary lengths of time. The following discussion of this effect will concentrate on parameter values of $\nu = 0.0122$ and $m_{cr} < m < 0.195$ (including the Earth–Moon–Sun parameter values).

At $m = m_{cr}$ the vectors along the stable and unstable manifolds are anti-parallel. As m increases from m_{cr} the angle between them decreases and they pass through 90° at $m \sim 0.05$. The angle between them comes to a minimum of $\sim 30^\circ$ at $m \sim 0.095$ before it begins to increase, reaching $\sim 50^\circ$ at $m = 0.195$. The minimum value of angle between the manifolds roughly corresponds to an observed change in the long-term properties along the stable and unstable manifolds, as will be discussed later. The angle is plotted in Figure 8.

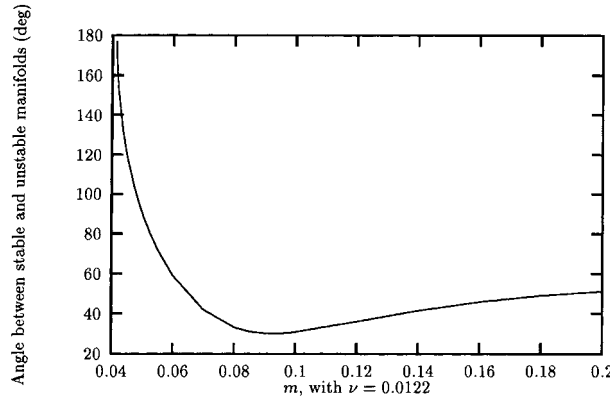


Figure 8. Angle between the period II orbit stable and unstable manifolds as a function of m for $\nu = 0.0122$.

The characteristic exponents of the unstable orbits start at $\lambda = -1$ at m_{cr} and monotonically change to -1.7 , -0.59 at $m = 0.195$. The frequency of the stable modes of this orbit, as measured by the angle along the unit circle, varies monotonically from 0.31π at m_{cr} to 0.35π at $m = 0.195$, notably passing through the frequency $\pi/3$ at $m = 0.10538$, which could potentially spawn a doubly unstable periodic orbit family of period 6π . Again, the long-term properties of the stable and unstable manifolds change shortly after the family passes through this particular parameter value.

The long-term dynamics of the stable and unstable manifolds have been studied numerically by integrating trajectories along them, initially starting close to the fixed-points defined by the π period family of orbits. For $m_{\text{cr}} < m < 0.11$ (including the value of m for the Earth–Moon–Sun system) the manifolds of these orbits appear to be trapped into a stable configuration in phase space. Due to the absence of a Jacobi integral for this four-dimensional dynamic system there is no such thing as KAM stability, in that there are no isolating integrals which guarantee that the dynamics are trapped within a certain region of phase space. Despite this, there is evidence of some sort of trapping in phase space in that the manifolds remain trapped within a small region of phase space for long time spans. The final evolution of the manifolds, as time grows infinitely long, cannot be established numerically due to integration errors, however.

Figures 9–10 present plots of the stable and unstable manifolds of the period π orbits at the Earth–Moon–Sun parameter values. These manifolds are one-dimensional objects embedded in four-dimensional space and thus cannot be unambiguously represented. We present the results by projecting the one-dimensional manifolds into the two-dimensional planes of (x, y) and (x', y') , thus delineating the manifold into a trajectory portion and a velocity hodograph portion.

Figures 9 and 10 show the L_4 stable and unstable manifolds for the Earth–Moon–Sun case. At this scale the two manifolds cannot be distinguished from each other.

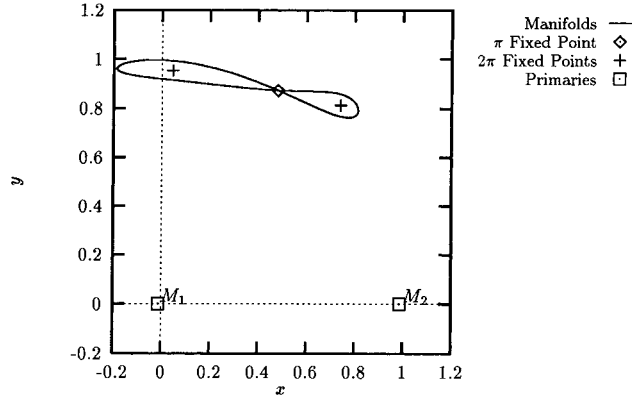


Figure 9. Stable and unstable manifolds for the Earth–Moon–Sun case ($\nu = 0.0122$, $m = 0.0808$): (x, y) coordinates. The plot shows one orbit along the stable manifold and one orbit along the unstable manifold.

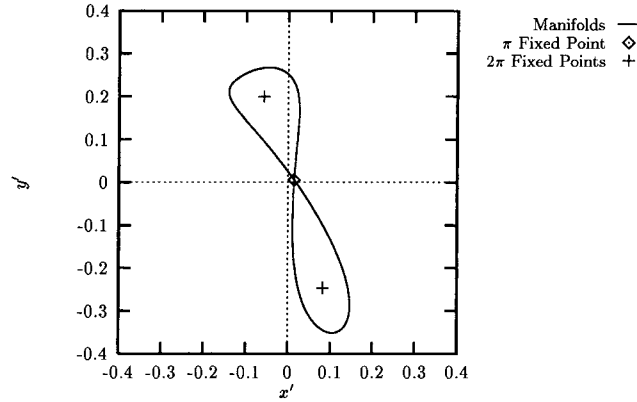


Figure 10. Stable and unstable manifolds for the Earth–Moon–Sun case ($\nu = 0.0122$, $m = 0.0808$): (x', y') coordinates. The plot shows one orbit along the stable manifold and one orbit along the unstable manifold.

Also plotted in the figure are the two fixed points associated with the stable 2π periodic orbit for the Earth–Moon–Sun case. Their proximity in phase space suggests some relation between the boundedness of the period π manifolds and the stability of the 2π period orbit.

Figures 11 and 12 present a detailed view of the manifolds in the neighborhood of the period π fixed point. The lines correspond to Poincaré maps every 2π time step. The unstable manifolds lie on the ‘outside’ of the structure while the stable manifolds lie on the ‘inside’ of the structure. Under iterations at a time step of π the stable and unstable manifolds jump from side to side. Note that all the unstable and stable manifolds points belong to a single orbit, respectively.

As m is increased up to larger values the bounded nature of the manifolds begins to fall apart. Starting around a value of $m \sim 0.11$ long-term integrations show that

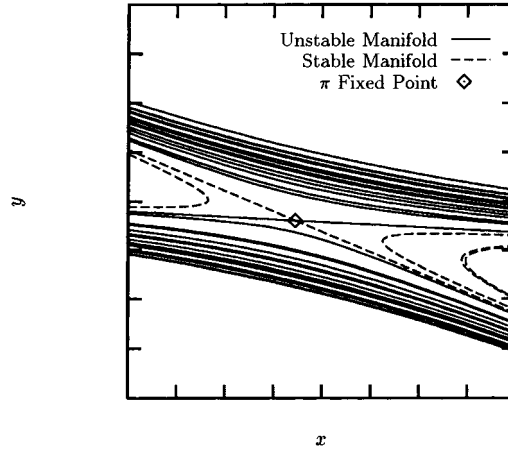


Figure 11. Detail of stable and unstable manifolds for the Earth–Moon–Sun case ($\nu = 0.0122, m = 0.0808$): (x, y) coordinates. The plot shows one orbit along the stable manifold and one orbit along the unstable manifold.

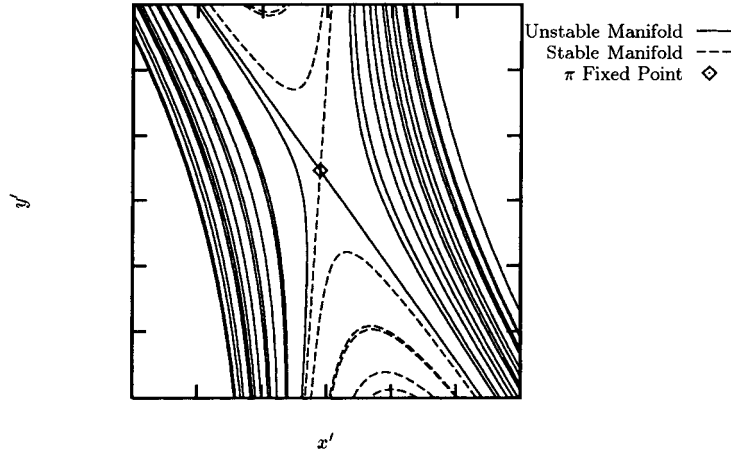


Figure 12. Detail of stable and unstable manifolds for the Earth–Moon–Sun case ($\nu = 0.0122, m = 0.0808$): (x', y') coordinates. The plot shows one orbit along the stable manifold and one orbit along the unstable manifold.

the regularity of the manifolds begins to break down and the manifolds eventually escape from the bounded area of phase space. This may also be the case for smaller values of m , but the time scale for escape is too large to be practically accessible with our numerical computations. By the time that $m = 0.12$, the manifolds rapidly depart from the structure and escape the system. For larger values of m the previously stable structure appears only briefly, if at all, as a transient structure.

It would be nice to relate the break down in the manifold stability with some other phenomena in the phase space, but that is difficult to do for such a system. The point at which the change in the manifold structure occurs is not immediately evident, as

it is a long-term phenomena and is not directly computed but is only observed. Also, there are several potential influences that may alter the structure, none of which can immediately be identified as the culprit, although there are some that we can rule out based on a plausibility argument.

For instance, the stability transition of the 2π orbit does not seem to be related to the change in manifold behavior. First, the critical value of m for the stability change does not occur until $m \sim 0.145$, at a larger value of m than where the change in manifold behavior was first seen. Also, the neighborhood of the unstable 2π orbits tends to be stable, due to the close proximity of a stable 4π orbit which traps the manifolds in a similar way that the π manifolds are trapped. A second possible clue to the change in the manifold structure may be seen in the changing angle between the stable and unstable manifold vectors in the neighborhood of the fixed point π . We noted before that the angle between these directions went through a minimum at a value of $m \sim 0.095$. Although the mechanism for how that might affect the observed loss of stability for the manifolds is not clear, the change may indicate some change in the topology of the phase space which influences how these manifolds may be trapped. A third possibility, seemingly unrelated to the previous two, involves the possible intersection of the period π orbit with a doubly unstable orbit of a higher period. As noted previously, at a value of $m = 0.10538$ the stable root of the period π family passes through an angle of $\pi/3$, which indicates that the current family may intersect with a family of period 6π . If such an intersection occurs, the 6π period family would be doubly-unstable, as the current family remains stable. The existence of this family has not been established as of yet. In this situation, the stable and unstable manifolds of that orbit may influence or even intersect with the manifolds of the π orbit.

Finally, applying transformation (67) and the time reversal to the above results we find that the unstable and stable manifolds associated with the L_4 fixed point become the stable and unstable manifolds, respectively, associated with the L_5 fixed point. Thus, all the properties mentioned about the L_4 π fixed point and manifolds apply to the L_5 π fixed point and manifolds, except for a shift in geometry and a reversal of the stable and unstable manifolds.

7.5. COMPARISONS WITH PREVIOUS RESULTS

Due to the simplified nature of our model, exact numerical comparisons with previous analyses do not make sense. However, if a qualitative agreement in results are found then our current model is shown to be useful as it provides a concisely stated model with which to study the dynamics of a small mass in the Earth–Moon–Sun system, or other such generalizations. If there is such a qualitative agreement then we feel that our model contains the essential difficulties of the problem, additional effects being of a perturbative nature.

In their analytical papers on motions near the Earth–Moon–Sun triangular points Schechter [6] and Kamel and Breakwell [3] both predict the existence of an unstable,

small periodic orbit located near the triangular points. Kolenkiewicz and Carpenter [4] perform a numerical analysis and find a similar orbit. All of these models contain the distance of the Sun from the Earth–Moon center of mass as an additional parameter, one we have eliminated by application of the Hill approximation. Due to this parameter their equations of motion have a one month periodicity (2π in our notation) and hence the small orbit has a one month period. This seems to contradict our result of a family of orbits with period π (half-month period), but on closer inspection there is not a problem as the unstable orbits found by the above researchers are of the form of two small ellipses of similar size to our π family of orbits, each circuited in a half-month period, thus indicating that this orbit is a bifurcation of a half-month period orbit as the length parameter is introduced. The small size of this parameter was our justification for its neglect in the Hill approximation, and we see that its inclusion, while changing the results, does not seem to have a major influence. Furthermore, if the Variation orbit is used to describe the relative motion of the primaries, it is not consistent to retain this parameter in the forces acting on M_3 , as the Variation orbit is defined without this parameter present. None of these researchers investigated the stable and unstable manifolds of these orbits.

In the above-mentioned papers the authors also discussed a stable periodic orbit with a one month period. Specifically we see that they found two very similar stable orbits, with their positions 180° out-of-phase with each other. We attribute the presence of these two orbits with the additional parameter kept by these researchers as discussed above. Again, while important, we see that its presence is not crucial to the basic dynamics occurring and, depending on the assumed motion of the primaries, may not be strictly proper to include this effect. In comparing the size and orientation of our 2π family with the orbits predicted and computed by the authors we see a very good general agreement. Our model allows the interesting addition of tracing this family as m decreases, indicating the connection this family has with the restricted three-body problem.

In the paper by Kamel and Breakwell [3] they discuss the possible consequences of including the eccentricity of the Earth–Moon relative motion in the standard Variation orbit. This would result in new periodic terms in the equations of motion with period incommensurate with the simple π periodicity in our model. Under such an additional perturbation we would expect that the periodic orbits found in our model would bifurcate into a quasi-periodic motion and that the stable and unstable manifolds would take on a more complicated structure. These are interesting and crucial questions we have not addressed here. It is expected that for a small eccentricity there would not be fundamental changes in some of the properties found here, but for larger eccentricity values some additional effects would probably be seen. Our model is general enough to allow for inclusion of these additional effects, as they just correspond to a more sophisticated solution to the Hill equation of relative motion, Equation (41).

8. Conclusions

The Restricted Hill four-Body problem was derived in this paper and applied to and analyzed for parameter values in the neighborhood of the Earth–Moon–Sun system. The model developed here is interesting in that it is a generalization of both the restricted three-body problem and the Hill problem, both of which may be recovered by setting the appropriate parameter value to zero. The two parameter values used are the mass ratio of one primary to the other (the ν parameter in the restricted three-body problem), and the parameter m from the Hill problem, geometrically related to the distance of the central body from the center of mass of the system of bodies of interest. The model as stated here is idealized in that it uses the Hill Variation orbit for the motion of the Earth–Moon system, although that can be changed by substituting a more general solution of the Hill problem.

An application of this model was also made, studying the orbit families related to the geometrical location of the L_4 and L_5 points in the three-body problem for parameter values in the neighborhood of the Earth–Moon–Sun system. Two periodic orbit families were investigated in detail, and the stable and unstable manifolds of one of these orbits was also investigated. These manifolds have an interesting property in that they accumulate onto a simple geometrical subset of phase space for parameter values in the vicinity of the Earth–Moon–Sun system. For larger values of the m parameter, the manifolds break out of this phase space subset and escape from the primaries.

Appendix

Following are the Fourier coefficients of the Variation orbit, to the order m^6 [7]:

$$a_0 = m^{2/3} \left(1 - \frac{2}{3}m + \frac{7}{18}m^2 - \frac{4}{81}m^3 + \dots \right), \quad (73)$$

$$\frac{a_1}{a_0} = \frac{3}{16}m^2 + \frac{1}{2}m^3 + \frac{7}{12}m^4 + \frac{11}{36}m^5 - \frac{30749}{110592}m^6 - \dots, \quad (74)$$

$$\frac{a_{-1}}{a_0} = -\frac{19}{16}m^2 - \frac{5}{3}m^3 - \frac{43}{36}m^4 - \frac{14}{27}m^5 - \frac{7381}{82944}m^6 + \dots, \quad (75)$$

$$\frac{a_2}{a_0} = \frac{25}{256}m^4 + \frac{803}{1920}m^5 + \frac{6109}{7200}m^6 + \dots, \quad (76)$$

$$\frac{a_{-2}}{a_0} = 0m^4 + \frac{23}{640}m^5 + \frac{299}{2400}m^6 + \dots, \quad (77)$$

$$\frac{a_3}{a_0} = \frac{833}{12288}m^6 + \dots, \quad (78)$$

$$\frac{a_{-3}}{a_0} = \frac{1}{192}m^6 + \dots \quad (79)$$

and

$$\frac{a_{\pm n}}{a_0} < \mathcal{O}(m^6), \quad (80)$$

$$|n| > 3. \quad (81)$$

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